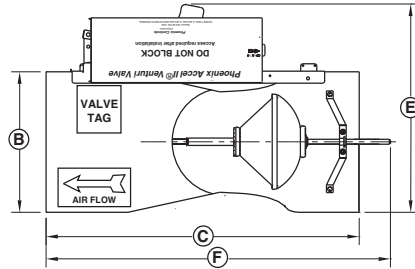
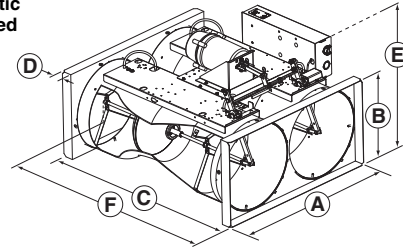


### DIMENSIONS AND WEIGHTS

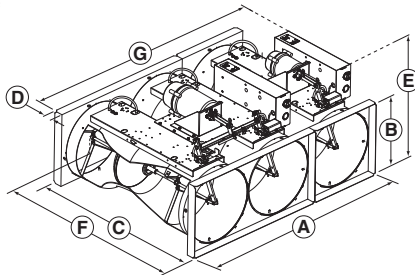
**Single**  
(EXV-electric)



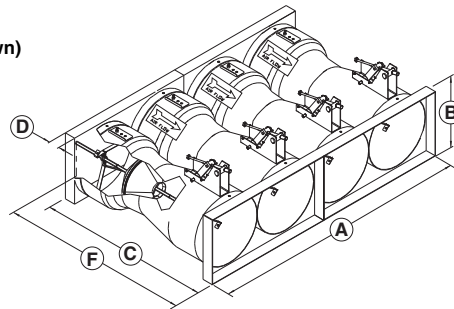
**Dual**  
(MAV-pneumatic  
normally closed  
shown)



**Triple**  
(MAV-pneumatic  
normally closed  
shown)



**Quad**  
(CVV shown)



|              | A*    |      | B*    |     | C     |     | D   |    | E**   |     | F     |     | G     |      | Weight<br>(CVV valves) |      | Weight<br>(all others) |      |
|--------------|-------|------|-------|-----|-------|-----|-----|----|-------|-----|-------|-----|-------|------|------------------------|------|------------------------|------|
|              | in.   | mm   | in.   | mm  | in.   | mm  | in. | mm | in.   | mm  | in.   | mm  | in.   | mm   | lbs                    | kg   | lbs                    | kg   |
| <b>8</b>     | —     | —    | 7.88  | 200 | 23.50 | 597 | —   | —  | 14.13 | 359 | 28.00 | 711 | 10.13 | 257  | 7                      | 3.2  | 12                     | 5.5  |
| <b>10</b>    | —     | —    | 9.67  | 246 | 21.75 | 552 | —   | —  | 16.13 | 410 | 26.20 | 665 | 11.20 | 284  | 7                      | 3.2  | 13                     | 6.0  |
| <b>12</b>    | —     | —    | 11.84 | 301 | 26.81 | 681 | —   | —  | 18.13 | 461 | 32.56 | 827 | 12.13 | 308  | 9                      | 4.1  | 16                     | 7.3  |
| <b>14</b>    | —     | —    | 13.88 | 353 | 30.00 | 762 | —   | —  | 21.43 | 544 | 38.09 | 968 | 13.52 | 343  | 13                     | 6.0  | 20                     | 9.1  |
| <b>2-10</b>  | 20.13 | 511  | 10.13 | 257 | 24.75 | 629 | 1.5 | 38 | 16.77 | 426 | 27.70 | 704 | 21.52 | 547  | 18                     | 8.2  | 30                     | 13.6 |
| <b>2-12</b>  | 24.13 | 613  | 12.13 | 308 | 29.81 | 757 | 1.5 | 38 | 18.75 | 476 | 34.60 | 879 | 24.76 | 629  | 23                     | 10.4 | 36                     | 16.3 |
| <b>3-12</b>  | 37.06 | 941  | 12.13 | 308 | 29.81 | 757 | 1.5 | 38 | 18.75 | 476 | 34.60 | 879 | 36.77 | 934  | 32                     | 14.5 | 52                     | 23.6 |
| <b>4-12</b>  | 48.26 | 1226 | 12.13 | 308 | 29.81 | 757 | 1.5 | 38 | 18.75 | 476 | 34.60 | 879 | 48.65 | 1236 | 46                     | 20.9 | 72                     | 32.7 |
| <b>2-14</b>  | 30.00 | 762  | 15.00 | 381 | 33.00 | 838 | 1.5 | 38 | 21.43 | 544 | 38.09 | 968 | 28.90 | 734  | 37                     | 16.8 | 50                     | 22.7 |
| <b>3-14*</b> | 45.00 | 1143 | 15.00 | 381 | 33.00 | 838 | 1.5 | 38 | 21.43 | 544 | 38.09 | 874 | 42.27 | 1074 | 40                     | 18.2 | 70                     | 31.8 |
| <b>4-14*</b> | 60.00 | 1524 | 15.00 | 381 | 33.00 | 838 | 1.5 | 38 | 21.43 | 544 | 38.09 | 874 | 57.13 | 1452 | 74                     | 33.6 | 100                    | 45.4 |

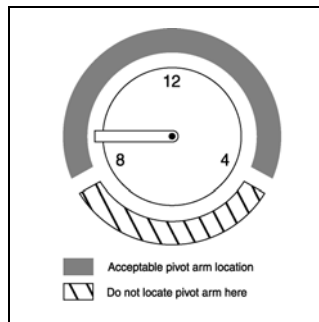
\*outer dimension    \*\*maximum of all valve types (some configurations may be smaller)

#### NOTES:

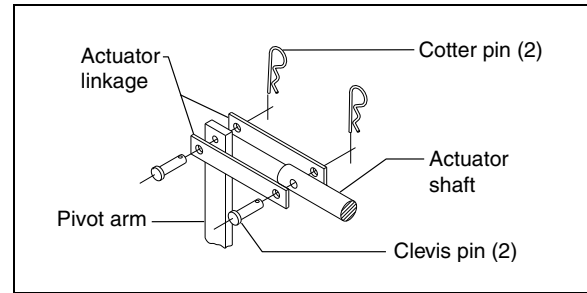
- Leave 14" (356 mm) access space to all electronic controls.
- Single valve circular flange dimensions can be found in the *Flanges for Single Body Valves* Product Data Sheet.
- Dimensions given are accurate to  $\pm 0.13"$  (3 mm).
- Triple and quad valve units are shipped as single and dual valves. Field assembly is required.
- Slip type flange material is 20 ga. galvaneal with powder coating.
- Valves need no additional straight runs before or after valve. However, as identified by dimension F above, the shaft needs up to 6.75" (171 mm) of unobstructed space in the duct on the inlet side of the valve in the maximum flow position.
- Weights given are approximate and listed for reference only. For shipping, add 6 lbs. (2.7 kg) for singles and duals; 12 lbs. (5.4 kg) for triples and quads.
- Linkages for NO valves and some NC valves must be connected by installer (pneumatic valves).
- "E" clip supplied for CVV adjustment mechanism. Must be inserted by installer (see Figure 2 on the next page).
- Dimensions do not account for valves with 3/8" (9.5 mm) insulation.
- Refer to the installation instructions on the next page for additional details.
- The 14-inch valve is not available with a pneumatic actuator at this time.

## INSTALLATION

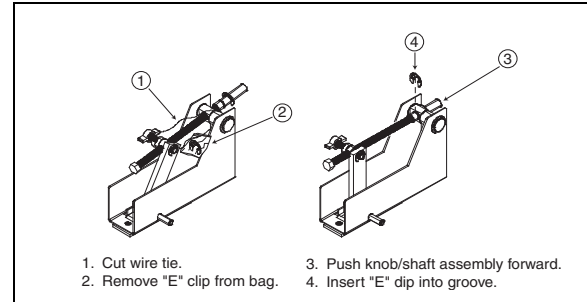
1. Read all instructions completely prior to installing valves.
2. Check that tag number on valve label matches HVAC schedule.
3. Verify correct airflow direction and orientation of the valve in the ductwork (e.g., horizontal). NOTE: Valves mounted out of horizontal or vertical position (as determined by a level) will affect valve performance.
4. Allow a minimum of 14 in. (356 mm) of free unobstructed space around the valve for access. In general, the valve may be installed in a 360° rotation. However, single body horizontal hood valves should be installed so that the pivot arm (see Fig. 1 and 4) is located between 8 and 4 o'clock (not within 4 to 8 o'clock).
5. Allow 6.75" (171 mm) of unobstructed space in the duct on the valve's inlet side for the shaft to reach the maximum flow position.
6. Linkage Connection—Many valves require field connection of pivot arm (see Figs. 1 and 2).
7. For multiple valves, assemble the valve sets on the floor using duct sealant and fasteners (see Fig. 3). Slip L style mating piece onto valve flange.
8. Use duct sealant on all valve/duct connections (or flange gaskets for circular flanges).
9. Install a hanger stock to support the ductwork within 12 in. (305 mm) of the valve connection. Install valve by sleeving it into the duct after the hanger stock is in place.
10. Follow the appropriate installation diagram (see Fig. 5). NOTE: Screws, fasteners, duct sealant, hanger stocks, companion flanges, and gaskets are not provided by Phoenix Controls.



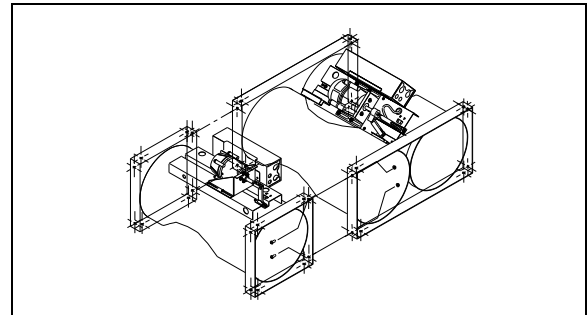
**Fig. 4** Pivot arm orientation installed.



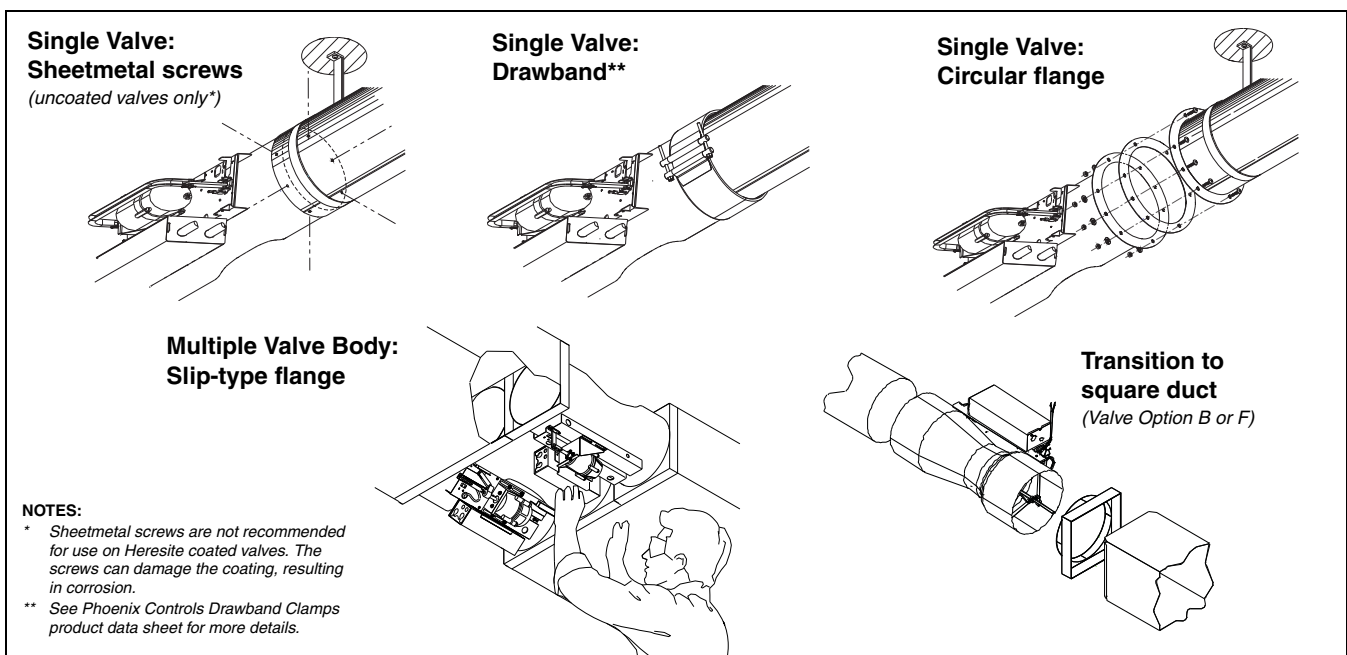
**Fig. 1** Linkage connection for normally closed pneumatic valves (normally open valves similar).



**Fig. 2** Constant volume valves.



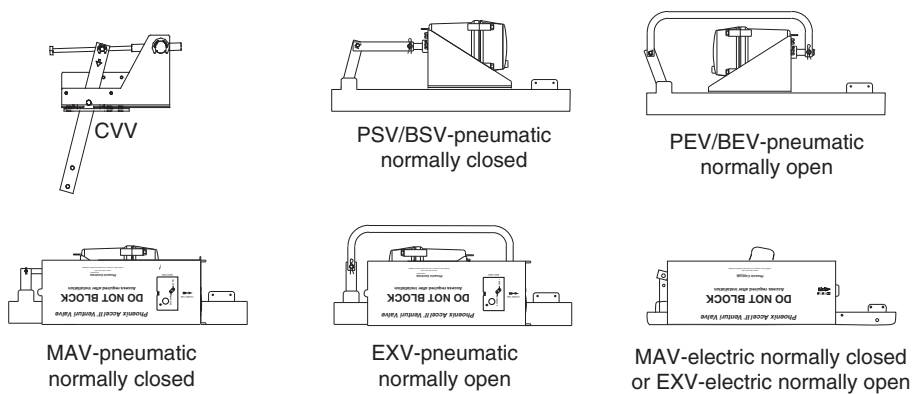
**Fig. 3** Points for fastening multiple assemblies.



**Fig. 5** Valve installation methods.

## VALVE FAMILY CONTROL TYPES

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## MAINTENANCE

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Accel II valves require no ongoing preventive maintenance. Once the field engineer has completed the field startup, the valves will provide years of continuous operation.